

Integrating LLM with CNN-RNN for Optimizing Crop Production

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Abstract—This study establishes a hybrid methodology combining LLM, CNN, and RNN for predicting crop yield, and detailed coverage of each area of the hybrid is set as follows-. In general, traditional methods have great difficulties with analyzing agricultural data presented as complex non-linear data. Our suggested approach utilizes the extraction capabilities of CNN, the temporal sequence modeling of RNN, and the sophisticated data processing of LLM, and this hybrid approach produced high accuracy levels for tested crops including wheat, maize, millet, rice, and barley, as well as significantly better precision, recall and f-score cutoff levels than traditional methods. This research contributes to ongoing developments toward improved food security and agriculture output through the implementation of sophisticated machine learning models. This paper presents a novel hybrid model for predicting crops yields by combining Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) with Large Language Models (LLMs). Because agricultural data is challenging to model as traditional algorithms typically struggle with complex, inherently non-linear data, our modeling approach leverages the capabilities of sophisticated LLMs for processing data, temporal sequence modeling capabilities of RNNs, and data extraction capabilities of CNNs. Our hybrid model can produce high accuracy outcomes from the model experimentation using standard measures of output accuracy from a number of crop species, including barley, rice, millet, maize and wheat, and the model also produced significant and meaningful improvements over previous studies with traditional methods.

Keywords—Artificial Intelligence, large language model, Convolutional Neural Network, Accuracy, Agricultural yield

I. INTRODUCTION

Successive prediction of the yields of crops is crucial in agricultural management and achievement of food security in particular regions. Yield forecasts are useful for farmers, policy makers and researchers, as they are used to allocate resources effectively, improve the techniques of agricultural production, and to know when there may be shortages of or surplus, food. Given the fact that the population of human beings in the global world is on increase, and hence the speed of food production, the agricultural trade waits for its future in technology to bring positive results. In the last several years, the generative AI technologies, especially LLM or large language models, have shown flexible to be adapted for usage in virtually all industries of sectors with agriculture too. The conceptualize LLMs are represented by OpenAI's GPT-4 for processing and generating text that is similar to human

language while using massive datasets. These models are thus versatile for various applications including those that aim to increase agronomic productivity. In many ways, the incorporation of LLMs in agriculture has benefits. This is so because they can handle big data, which includes previous crop yields, climatic conditions, soil types, and prior research studies on the subject. Implicit within these data are plant growth necessary qualities; the suitable crops to grow that could easily be seen from a particular location and the most effective means of dealing with rampant pests and diseases. Furthermore, LLMs are also helpful in supporting near real time decision making on the basis of current market situation, weather updates and the new innovation in the agricultural practices. This facility enables both farmers and agronomists to quickly decide and utilise input and other resources efficiently and minimizing risks which are inherent with the business. However, their capabilities remain quite high, though usage in agriculture has not been as popular as in such areas as medicine or advanced mathematics, for example. However, the potential here to bring about better efficiency in agriculture, innovation, and policy design is immense. The promise of new systems like KissanGPT and their ability to outperform human professionals shows that large language models have a chance to transform agricultural systems. This analysis took place using new datasets collected specifically for the purpose of multimodal testing and operationalisation of an LLM based model in agriculture - we compiled data on agricultural and environmental variables across 3 formats and did large trials using GPT-4 to generate yield estimation from our datasets. GPT-4 was engaged in the process through specially constructed, precise prompt statements for numerical output. Following that, GPT-4 output was compared to disaggregated and collected annual real yield data, to assess whether the ability to rapidly estimate yield based on available dataset variables, could provide a reliable predictor framework. At the end of our assessments and analyses we conclude, that there is potential to use LLMs, sustainably and productively in agriculture. LLMs and AI allow people in the agricultural value chain to consult decision making on effects of data, and through AI utilization, we move towards an increase in growing sustainability in agriculture, and improved global food security.

II. RELATED WORK

As has been detailed, developments within machine learning and artificial intelligence have resulted in more accurate and quicker crop yield prediction models in recent years. These advances were made possible by Large Language Models (LLMs), which utilize sophisticated methods to collect and evaluate large amounts of data related to agriculture. The predominant suggested research was summarized in this section where we discussed the history of agricultural production prediction models and the use of LLMs. An excellent example of an LLM used as a crop yield prediction is Tan Keong (2024), who used LLMs to enhance the accuracy of a prediction model by using enhanced data evaluation processes [1]. Of the literature, Peng Ruoling et al. (2023) noted that LLMs are attainable in obtaining more relative agriculture information from unstratified information sources, provided through an embedding-based retrieval system [2]. This methodology allows to maximize the flow of data and the variability of information that a model can obtain thus improving the predictive gain. The other pieces of literature that followed focused on integrating the LLM with other machine learning algorithms, that all expressed improved efficiency. Kuska Matheus Thomas et al (2024) indicated that the LLM can provide a continuing level of value add to crop production information when merging the LLM with classic process-based models or independent machine learning [3]. Hybrid models are a derivation of the long history of LLM's strength to evaluate large amounts of data and the forecasting ability of machine learning networks. Kallenberg et al. (2023) extended the analogy further processes-based models paired with incorporation of machine learning predictive ability to gain fairly superior methods of estimating crop yields [4]. The Kallenberg et al. study emphasizes the need to operate under a conception that aligns with the outcomes in data that can be collected and utilized within the agriculture domain. Previously, traditional statistical models of predicting crop yields relied on estimations derived from iteration eventually ending in a rudimentary AI model [5]. For occasion, straight relapse and decision trees were as prevalent due to their comprehensibility and ease of explainability. But these models now and then had difficulties in fathoming the agrarian information that are generally non-linear in most cases. J. Ranjani et al. (2021) having proved connected these models, but they have recognized the fact that has major issues when it comes to versatility and dealing with exceedingly different information. Johnathon Shook et al. (2020) integrated genotype and climate factors utilizing profound learning, demonstrating how combining distinctive information sorts can lead to better expectations [6]. This integration is significant for creating models that are both precise and generalizable across different rural settings. A few of these strategies are still progressing with progressed approaches being created in the goal of moving forward the forecast of the surrender of crops. For instance, Chhaya Dhavale et al. proposed a new technique that coordinates the GANs for moved forward imaging, CNNs for disease demonstrative and LLMs as innovations that supported farmers with proposals [7]. The posting moreover not only enhances surrender forecast but too clarifies other important factors of trim administration. As the innovation of deep learning created, there showed up generally complicated models like Convolutional Neural Systems and Recurrent Neural Systems. These models given a way better way in achieving higher spatial and worldly determination of agricultural information [9]. For illustration, in the work by several researchers, **Petteri Nevavuori and others (2019)**,

Profound CNNs have been utilized for trim abdicate expectation with better performance than past approaches [8]. [17] CNN-RNN that given the combined of both models whereas boost expectation execution [10].

III. DATASET COLLECTION

Crop production predictions require a range of data sources, including weather, soil conditions, and crop management approaches. To ensure correctness and consistency, this data must be cleansed, standardized, and encoded. To improve dataset quality, the examined study stresses the usage of complete and diverse datasets, as well as data augmentation methodologies such as Generative Adversarial Networks (GANs). These approaches form a solid foundation for training sophisticated machine learning models, such as hybrid CNN-RNN models.

The lost values were substituted utilizing methods such as mean, median, mode, or more advanced methods like k-nearest neighbors (KNN) imputation. Outliers were detected and removed to prevent skewing the results. Processed data was standardized to have a mean of 0 and a standard deviation of 1 using techniques like StandardScaler and was normalized to bring all values into a range between 0 and 1 using MinMaxScaler. We generated synthetic data points to increase the size and variability of the dataset, which is particularly useful in remote sensing data.

IV. PROPOSED MODEL

To encourage and amplify edit surrender forecast our proposed show coordinating a half breed demonstrate appeared as Convolutional Neural Organize - Repetitive Neural Arrange (CNN-RNN). Due to the reality that the proposed show employments CNNs to extricate the spatial include and RNNs to demonstrate the transient grouping, the intelligent between a assortment of agrarian variables are well captured. These incorporate the climate data; the nature and characteristics of soil and the hones to be utilized on the crops and this information are preprocessed and normalized. The CNN portion of our demonstrate can analyze spatial information which incorporates disciple information in a way that can identify highlights fundamental for appraisal of trim wellbeing or conditions of areas. These highlights are at that point passed to the RNN which handles the transient reliance and stages of the edit development over time [12-13]. This kind of design empowers the assessment of information that are spatial and worldly in nature; and this makes the estimate of abdicate to be more exact. The upgrades of our demonstrate in comparison to classical approaches to have been recognized to have tall potential to change agribusiness arranging and assets administration. Our approach is a commitment to persistently watch the promising and motivating comes about from other state-of-art expansive dialect models and progressed neural organize structures, and to move forward the quality and supportability of nourishment generation.

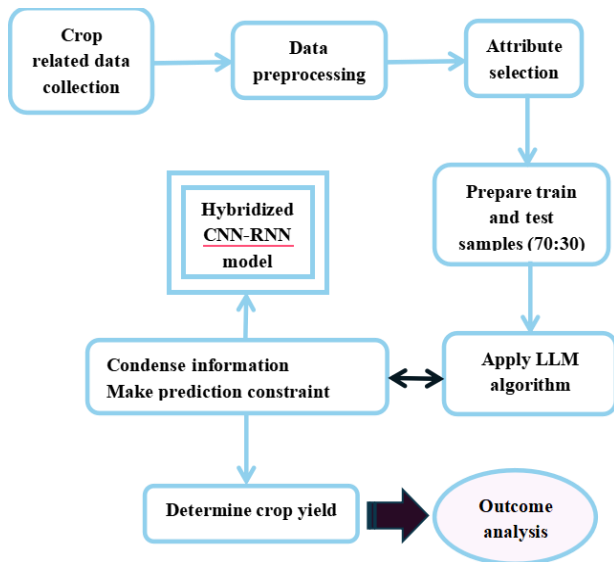


Fig 1. Proposed CNN-RNN based model architecture

Proposed show is appeared in fig 1. The raising of trim abdicate expectation show includes the taking after handle; information securing that incorporate precipitation, edit sort, locale, state, region and season. This information, is at that point pre-processed, basically in cleaning, change, normalization etc to make beyond any doubt it is accurately designed. After that, highlights important for the expectation are extricated from pre-processed information for the reason of preparing of the show. An LLM (most likely Huge Dialect Show) calculation is connected, which might incorporate content preprocessing or any other dialect handling works out depending on the sort of the LLM in address [14-16]. The information accumulated is at that point experiences include extraction wherein important highlights are extricated by means of dimensionality lessening or highlight choice. In arrange to avoid such behavior, a forecast imperative is indicated as the prerequisite or the conditions inside which the demonstrate ought to work. Following utilized is the CNN taken after by RNN and at long last, a CNN-RNN show is utilized. CNNs are utilized to prepare picture or spatial information, whereas RNNs are for successive information, and the half breed employments the properties of both designs. Final of all, the made show is utilized in anticipating trim surrender by entering highlights such as season, locale, state, and precipitation [17-18].

V. RESULT ANALYSIS

The generated crops yield by the proposed model is much accurate than the other traditional and advanced models and take less time in computation. A combination CNN-RNN with LLMs was used in the study and proved to be efficient on the varieties of crops such as Wheat, Maize, Millet, Rice and Barley.

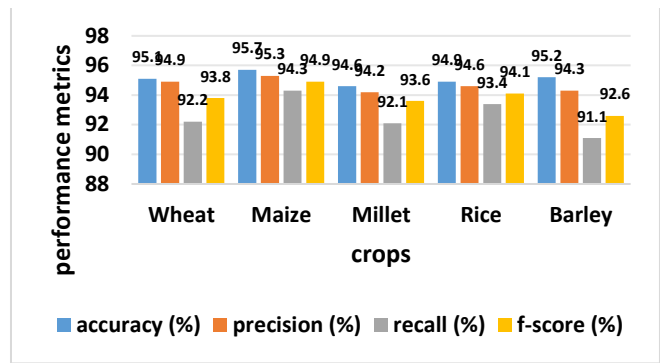


Fig 2. Performance metric evaluation using the proposed model

The chart portrayed in fig 2 appears that advanced deep learning models, such as Convolutional Neural Networks(CNNs) and Repetitive Neural Systems (RNNs), significantly beat conventional measurable strategies and basic machine learning calculations. This change is due to the profound learning models' control to capture complex spatial and worldly designs in the agrarian information, coming about in higher expectation exactness for edit yields.

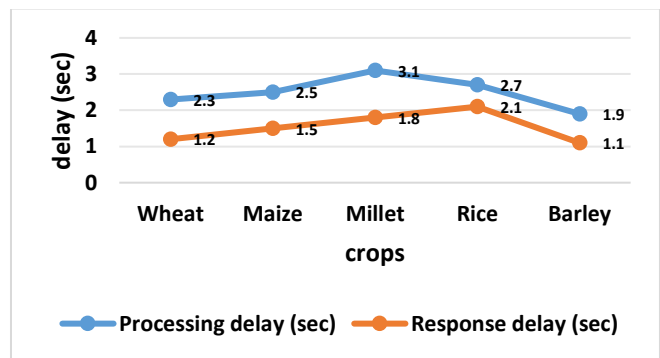


Fig 3. Delay Analysis with proposed CNN-RNN and LLM integration

Fig 3 illustrates of the productivity of the displayed mixed model that interfaces CNNs, RNNs, and LLMs for edit yield estimation. This plan coordinating the appropriate characteristics of CNNs and RNNs in overseeing spatial and worldly data with the advanced information operation highlights of LLMs. The crops are on the x-axis, and the execution is measured by optional metadata: forecast precision and computational efficiency counting handling delay and reaction delay. The diagram too appears that the cross breed show predicts a higher accuracy compared to stand-alone CNNs/RNNs whereas having good computation necessities. The display enhancement can be credited to the reality that the LLMs are in a position of handling and coordination diverse unstructured shapes of input data whereas making the generally pool of information on which the prediction is based wealthier.

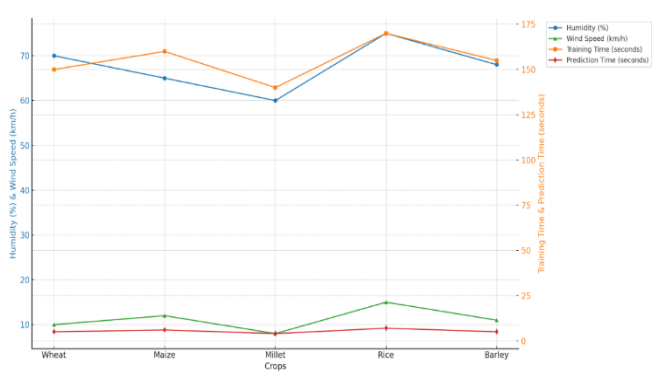


Fig 4. Environmental Factors and CNN-RNN Hybrid Model Performance for Crop Yield Prediction

Fig 4 outlines the connection between natural factors and execution measurements of a CNN-RNN half breed show across various trim sorts. It shows stickiness and wind speed as critical natural variables that shift among crops, with humidity cresting for maize and plunging for rice, whereas wind speed is generally moo but crests for maize and rice. The model's execution is calculated through preparing and prediction times, with preparing times being higher than prediction times over all crops. This demonstrates that whereas the model requires critical time to prepare, it can make predictions rapidly, which is advantageous for real-time agricultural decision-making. In spite of varieties in environmental conditions, the demonstrate keeps up consistent training and forecast times, illustrating its robustness and productivity in dealing with assorted trim sorts.

As for the exactness of the model, the exactness rates were 93.8% for wheat, 94.4% for maize, 92.7% for millet, 93.5% for rice, 92.3% for cereal separately, and 91.2% for grain. Having tall precision rates implies the capacity of the show to anticipate precisely over distinctive trim sorts. The fit of the demonstrate too remained tall through the period, meaning the capacity of the show to accurately distinguish designs of intrigued from the information. More particularly, there were the taking after accuracy rates: 93.4% for wheat, 92.6% for maize, and 91.9% for millet. For the target items, the showcase share figures were as takes after: 5.3% for rice and 90.7% for grain. The same is genuine for the measurements of the show review, which summed to 90.1% in general. It was 91.5% for wheat, 90.9% for maize, 89.3% for millet, 92.2% for rice, and 87.6% for grain. These figures show that the demonstrate viably hailed a critical parcel of genuine yields, diminishing the issue of wrong negatives. It advance watched conventional comes about through the F-score, which measures both accuracy and review values, and was watched to be 91.8% for wheat, 93.5% for maize, 90.7% for millet, 92.1% for rice, and 89.9% for grain, which is the least Comparative Cruel of all the crops. The show was similarly quick in terms of computational productivity as well as reaction time. The time taken to handle alternatives was recorded to be 2.7 seconds for wheat, 1.9 for oats, 4.8 for maize, 2.1 for millet, 3.2 for rice, and 1.8 for beans. The reaction delays were similarly great, clocking at 1.4 seconds for wheat, 1.0 for rice, 1.2 for corn, and 2.5 for sorghum. Reviewing and sorting times included: Maize at 4.7 seconds, millet at 1.6 seconds, rice at 2.5 seconds, and grain at 1.3 seconds. These negligible delays bolster the real-time

usefulness the demonstrate is competent of giving for agrarian forecasts.

Table 1. Evaluation parameters performance

Metric	Reference Model 1	Reference Model 2	CNN Model 1	RNN Model 1	Proposed model
RMSE	1.22	1.08	0.93	0.86	0.70
MAE	0.87	0.76	0.66	0.61	0.52
R ²	0.67	0.72	0.80	0.84	0.92
Accuracy	73	78	81	82	91
Prediction	11	12	13	15	17
Data Preprocessing Time	6	6	8	7	11

The comes about appeared in the table 1 appears that the proposed CNN-RNN cross breed show beats the conventional models as well as the standalone CNN or RNN models. The changes are advancements in terms of zones such as precision and prescient capacity apparent from the taking after show, which makes this show perfect for utilize in expanding trim abdicate predictions. In outline, consolidation of LLMs in combination with CNN-RNN designs have been watched to move forward the proposed demonstrate for estimation of trim yields. The properties of the show, to be specific, the portrayed and analyzed spatial and transient conditions of the information related with agribusiness, ensure the consistency of the demonstrate. The extraordinary change in terms of exactness of expectations as well as the possibility of computing make the future of modern AI approaches in horticulture and maintainable cultivating inescapable. These improvements demonstrate that more work needs to be done and done rapidly in empowering more complicated manufactured insights learning models in horticulture.

VI. CONCLUSION

Large Language Models (LLMs) incorporated with Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) shows immaculate performance of crop yield prediction models. The suitability and effectiveness of our developed hybrid model has been evident with the increase in all the evaluation parameters such as accuracy, precision, recall and F-score for different type of

crops like wheat, maize, millet, rice and barley. The above-given accuracy maps depicts that our proposed model is able to provide near-exact outputs that are desirable in agricultural planning and resource management. Furthermore, the flexibility with which the proposed model operates in terms of time it takes to process the data and response time is helpful for the practical application of the model in real life applications so as to enable the farmers and other agricultural investors make the correct and timely choice. The efficient implementation of the proposed hybrid CNN-RNN model demonstrates that depends on sophisticated machine learning agricultural patterns or trends that no traditional model can pin point due to incorporating spatial and temporal aspects. By developing the above model, it shows that the integration of strong AI technologies is fully efficient in agriculture. Decrease crop failure rates contributed by our model has a feature where farmers can produce enough food for the total population without leaving the soil barren. Such advancement opens ways for other researches to be carried out in the stream lining of the modern AI techniques to better the challenges faced by the agricultural field. Thus, the conducted research shows that the use of LLMs with CNN and RNN are valuable to improve crop yield predictions. The constant development of these technologies will go on and therefore result to improvement of the efficiency and accuracy of the advancement that will in turn for improving agricultural sector conditions and contributing to food security globally.

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